

Mohammad Nusairat

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EDUCATION

University of Illinois at Chicago

Chicago, IL

B.S. in Data Science with a Concentration in Computer Science, GPA: 3.50

May 2025

- Related Coursework: Advanced Data Structures & Algorithms, Intro to Machine Learning, Database Systems, Optimization for Analytics, Statistical Methods and Computing, Software Design, Computer Algorithms

University of Illinois Urbana-Champaign

Champaign, IL

M.S. in Computer Science, GPA: 4.0

May 2027

- Related Coursework: Applied Machine Learning, Cloud Computing Applications, AI Agents in the Wild, Introduction to Data Mining, Theory & Practice of Data Cleaning, Foundations of Data Curation

SKILLS

Languages: Python, SQL, Java, Javascript, TypeScript, R, C++, C, C#, SAS

Data & AI: PySpark, Pandas, Databricks, OpenAI API, MLflow, LlamaIndex, Azure AI Search, Great Expectations

Technologies: Azure (Blob, App Services, Key Vault), AWS (S3, Lambda, RDS, Glue), Flask, React, Docker, Git

EXPERIENCE

AbbVie

North Chicago, IL

AI Data Scientist II

September 2026 - Present

- Building AI solutions for clinical data operations and pharmaceutical innovation as part of the FAiM team.

Constellation Energy

Remote

AI Data Scientist I

August 2025 - September 2026

- Built an **AI-powered sales platform** leveraging **LLM-based reasoning**, **RAG**, and **heuristic pipelines**, generating **~\$30M in revenue** across PoC & MVP, **optimizing & automating the energy sales lifecycle**.
- Developed scalable **data science workflows**, **pipelines** and **jobs**, utilizing **Apache Spark**, **Kedro** & **Databricks** for **feature engineering**, **model development & evaluation**, and **interfacing** output tables through **Azure Data Factory**, populating **SQL staging tables** for **backend API integration**.
- Implemented **Pydantic structured I/O**, **PyTest** unit testing, **Great Expectations** data quality checks, **MLflow** tracing/evaluations, and **Azure OpenTelemetry** logging for pipeline **reliability & observability**.
- **Reduced deployment runtime by 71%** by packaging dependencies as Python wheels, **automating cluster installs** via shell scripts, and implementing **asynchronous API calls** with a **customized caching mechanism**.
- Oversaw **complete delivery of multiple features** (requirements gathering, refinement, development, iteration, and documentation), **receiving distinction from managers and directors** across the organization.
- Onboarded multiple senior data scientists and coached on **optimal pipeline & YAML parameter** design.
- Managed stage & prod **Azure Pipeline deployments**, debugging model/pipeline failures, data inconsistencies, and **LLM API batch processing** issues to ensure system reliability & **resolve edge-cases in large datasets**.
- Conducted **PR reviews**, **knowledge transfer** sessions, & **story development guidance**, cultivating strong engineering standards, resulting in **maintainable & scalable** repository patterns across the Data Science team.
- Only Data Scientist selected to **scale & replicate project success** across another department in Constellation.

PROJECTS

Arabic YouTube Transcript Translator

- Built a **React + Vite** frontend and **Dockerized Node.js + Express.js** backend web app that transcribes Arabic YouTube videos, translates them into English, and generates timestamped PDF, Markdown, and SRT files
- Streamlined content creation for my personal business by **reducing Arabic content preparation time by 86.7%** and **increasing transcription and translation accuracy by 20%** as tracked by internal QA logs
- Achieved **50k+ LinkedIn** views with **330+ users** and **16+ hours transcribed** on launch day

Travel Planning Agent

- Built a **ReAct** travel planning agent in **LangChain** generating multi-city itineraries under complex constraints
- Implemented **MCP** architecture with **20 tool integrations**, enabling **iterative reasoning** on structured data
- Designed **constraint class** temporal logic, budget limits, & dependency rules **ensuring valid itineraries & robust disruption handling** with **LLM-as-judge** scoring and **token/API tracking** for **benchmark** validation